

iii) Probability of at least two success:

$$\begin{aligned} & f(2) + f(3) + f(4) + f(5) + \dots \\ &= 1 - \{f(0) + f(1)\} = 1 - e^{-3} - 3e^{-3} = 1 - 4(0.0498) \end{aligned}$$

Example: In a textile mill, on an average, there are 5 defects per 10 square feet of cloth produced. If we assume a poisson distribution, what is the probability that a 15 square feet cloth will have atleast 6 defects?

Let X be a random variable denoting the number of defects in a 15 square feet piece of cloth.

Since on an average, there are 5 defects per 10 square feet of cloth, there will be on an average 7.5 defects per 15 square feet of cloth i.e., $\lambda = 7.5$. We are to find $P(X \geq 6) = 1 - P(X \leq 5)$.

With the help of Table 27.5, we obtain $P(X \leq 5) = .2415$.

Therefore, $P(X \geq 6) = 1 - .2415 = .7585$.

Table 27.5: Probability distribution for number of defects

X	P (X)
0	0.0006
1	0.0041
2	0.0156
3	0.0389
4	0.0729
5	0.1094

Proposition: If x_1 and x_2 are independent poisson distributions with parameters λ_1, λ_2 , respectively, then $(x_1 + x_2)$ also follows poisson distribution with parameters $(\lambda_1 + \lambda_2)$.

Suppose x_1 and x_2 are distributed independently in the Poisson form with parameter λ_1 and λ_2 respectively. The sum $x_1 + x_2$ can then take the values 0, 1, 2, ...

$$\text{Also, } P[x_1 + x_2 = k] = \sum_{k_1=0}^k P[x_1 = k_1]P[x_2 = k - k_1]$$

$$= \sum_{k_1=0}^k \frac{\exp[-\lambda_1] \lambda_1^{k_1}}{k_1!} \times \frac{\exp[-\lambda_2] \lambda_2^{k-k_1}}{(k-k_1)!}$$

$$= \frac{\exp[-\lambda_1 - \lambda_2]}{k!} \sum_{k_1=0}^k \binom{k}{k_1} \lambda_1^{k_1} \lambda_2^{k-k_1}$$

$$= \exp[-(\lambda_1 + \lambda_2)] \frac{(\lambda_1 + \lambda_2)^k}{k!},$$

which shows that $x_1 + x_2$ is itself a poisson variable with parameter $\lambda_1 + \lambda_2$. It immediately follows that if $x_1, x_2, \dots, x_n$ are independently distributed poisson variables with $\lambda_1, \lambda_2, \dots, \lambda_n$ then the sum $x_1 + x_2 + \dots + x_n$ is also a Poisson variable with parameter $\lambda_1 + \lambda_2 + \dots + \lambda_n$.

The above results give, in particular, the sampling distribution of the statistic $x_1 + x_2 + \dots + x_n$ when $x_1, x_2, \dots, x_n$ are a random sample from a Poisson distribution with parameter λ . This sampling distribution is also of the Poisson form with parameter $n\lambda$.

Check Your Progress 2

- 1) The mean and standard deviation of a binomial distribution is given by 4 and $\sqrt{\left(\frac{8}{3}\right)}$ respectively. Find the values of n and p .

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- 2) Prove that poisson distribution is a limiting case of binomial distribution.

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- 3) The overall percentage of failure in a certain examination is 40. What is the probability that out of a group of 6 candidates at least 4 passed examinations?

Hint: x follows a Binomial distribution and the event of a candidate passing the examination is a success. Thus, the probability of success:
 $f(x) = {}^6C_x (60/100)^x (40/100)^{6-x}$. Find out $f(4) + f(5) + f(6)$

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- 4) In 10 independent throws of a dice, the probability that an even number will appear 5 times is twice the probability that the even number will appear 4 times. Find the probability that an even number will not appear at all in 10 independent throws of the dice.

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- 5) In turning out some toys in a manufacturing process in a factory, the average number of defectives is 10%. What is the probability of getting exactly 3 defectives in a sample of 10 toys chosen at random ? Use poisson approximation to the binomial distribution (take $e=2.72$).

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- 6) 2% of the items made by a machine are defective. Find the probability that 3 or more items are defective in a sample of 100 items? ($e^{-1} = 0.368$, $e^{-2} = .135$, $e^{-3} = .0498$)

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- 7) Find the probability that 7 out of 10 persons will recover from a tropical disease if we can assume independence and the probability that anyone of them will recover from the disease is 0.8.

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27.4.3 Normal Distribution

The normal distribution, which we will study in this section, is the corner stone of modern statistical theory. It was first investigated in the eighteenth century when scientists observed an astonishing degree of regularity in errors of measurement. They found that the patterns of the errors of measurement could be closely approximated by continuous curves, which they referred as the “normal curves”. The mathematical properties of such continuous curves were first studied by Abraham de Moivre, Pierre Laplace and Karl Gauss.

A continuous random variable X is said to follow a normal distribution, and it is referred to as a normal random variable, if and only if its probability density is given by

$$f(x) = n(x, \mu, \sigma) = \frac{1}{\sigma\sqrt{2\pi}} \times e^{-\frac{1}{2}\{(x-\mu)/\sigma\}^2} \text{ for } -\infty < x < \infty, \text{ where } \sigma > 0$$

The shape of normal distribution is like a bell and is shown in Fig 27.2.

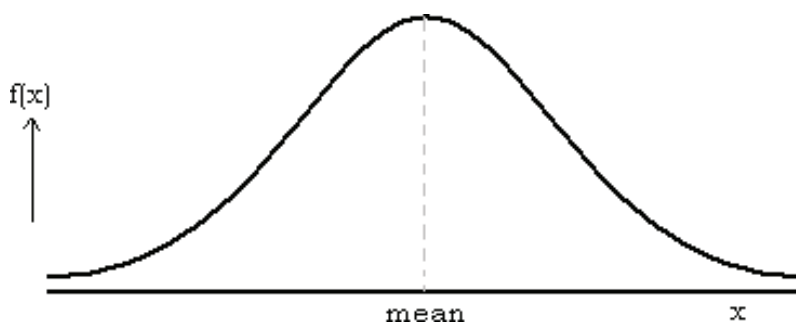


Fig. 27.2: Normal Distribution

While defining the p.d.f. of normal distribution we have used the standard notations, where σ stands for standard deviation and μ for mean of the random variable X . Note that $f(x)$ is positive as long as σ is positive which is guaranteed by the fact that standard deviation of a random variable is always positive. Since $f(x)$ is a p.d.f. while X can assume any real value, integrating $f(x)$ over $-\infty$ to ∞ we should get the value 1. In other words, the area under the curve must be equal to 1. Let us show that

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \times e^{\frac{1}{2}\{(x-\mu)/\sigma\}^2} dx = 1$$

We substitute $(x - \mu)/\sigma = z$ in the R.H.S of the above equation to get

$$\int_{-\infty}^{\infty} \frac{1}{\sigma\sqrt{2\pi}} \times e^{\frac{1}{2}\{(x-\mu)/\sigma\}^2} dx = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}z^2} dz = \frac{2}{\sqrt{2\pi}} \int_0^{\infty} e^{-\frac{1}{2}z^2} dz$$

[Since the p.d.f. is symmetrical, integrating the function from 0 to ∞ twice is the same as integrating the same function $-\infty$ to ∞]

$$= 2/\sqrt{2\pi} \times \sqrt{\pi}/\sqrt{2}$$

$$= 1$$

Normal distribution has many nice properties, which makes it amenable to be applied to many statistical as well as economic models.

Properties of Normal Distribution

- Normal distribution is a continuous probability distribution.
- Normal distribution has two parameters, mean and standard deviation given by μ and σ respectively.
- Normal distribution is symmetric around the mean μ , which is at the same time the mode and the median of the distribution
- As a corollary to the above property, we can say that the first and third quartiles are equidistant from the mean of the normal distribution. Approximately,

$$Q_1 = \mu - 0.67 \times \sigma$$

and

$$Q_3 = \mu + 0.67 \times \sigma$$

- All odd order central moments of the normal distribution are 0.

In general $\mu_{2r} = 0$ for $r = 1, 2, 3, \dots, \mu_{2r-1}$

- The two tails of the distribution are extended to infinity on both sides of the mean. The tails of the distribution never meet the horizontal axis. The maximum ordinate of the p.d.f is at the mean, which is given by $1/\sigma\sqrt{2\pi}$.
- However, beyond the points $x = \mu - 3\sigma$ and $x = \mu + 3\sigma$ the curve is close to the axis, that is the area under the curve beyond the two points in both

directions can be taken as zero.

- The points of inflexion (where the second derivate of $f(x)$ is zero and changes sign), of the normal curve are located one standard deviation away from the mean, namely at $x = \mu + \sigma$ and $x = \mu - \sigma$ respectively. At these two points, normal curve changes its curvature.
- It is unimodal: its first derivative is positive for $x < \mu$, negative for $x > \mu$, and zero only at $x = \mu$.
- The probability that a normal random variable X equals any particular value is 0.
- The probability that X is greater than a equals the area under the normal curve bounded by a and plus infinity.
- The probability that X is less than a equals the area under the normal curve bounded by a and minus infinity.
- Additionally, every normal curve (regardless of its mean or standard deviation) conforms to the following “rule”.
 - 1) About 68% of the area under the curve falls within 1 standard deviation of the mean.
 - 2) About 95% of the area under the curve falls within 2 standard deviations of the mean.
 - 3) About 99.7% of the area under the curve falls within 3 standard deviations of the mean.
- If a random variable X follows normal distribution with the mean and variance μ and σ respectively, then the random variable $z = (x - \mu)/\sigma$ is called standard normal variable. It has a density function:

$$f(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz \dots \dots \dots -\infty < z < \infty$$

- The continuous probability distribution defined above is known as standard normal distribution. In fact, this is a special kind of probability distribution with mean zero and standard deviation 1.
- If X and Y are two normal variables with mean μ_1 and μ_2 and standard deviation σ_1 and σ_2 , then $(X + Y)$ is also a normal variable with mean $(\mu_1 + \mu_2)$ and variance $(\sigma_1^2 + \sigma_2^2)$.
- The moment generating function of a normal curve is given by

$$M_X(t) = e^{\mu t + 1/2 \cdot \sigma^2 t^2}$$

$$M_X(t) = \int_{-\infty}^{\infty} e^{tx} \times \frac{1}{\sigma\sqrt{2\pi}} e^{-1/2 \left(\frac{x-\mu}{\sigma} \right)^2} dx$$

The above expression could be written, after some algebraic manipulation as the following:

$$M_X(t) = e^{\mu t + 1/2(t\sigma)^2} \times \frac{1}{\sigma\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-1/2 \left[\frac{x - (\mu + t\sigma^2)}{\sigma} \right]^2} dx = e^{\mu t + 1/2(t\sigma)^2}.$$

$$[\text{since } 1/\sigma\sqrt{2\pi} \int_{-\infty}^{\infty} e^{-1/2 \{ [x - (\mu + t\sigma^2)] / \sigma \}^2} dx = 1]$$

Differentiating $M_X(t)$ with respect to t twice, we can get

$$M'_X(t) = (\mu + \sigma^2 t) M_X(t)$$

$$M''_X(t) = [(\mu + \sigma^2 t)^2 + \sigma^2] M_X(t)$$

Substituting $t = 0$ in the above two equations

$$M'_X(0) = \mu$$

$$M''_X(0) = \mu^2 + \sigma^2. \text{ Therefore, } E(X) = \mu \text{ and Variance } (X) = \sigma^2.$$

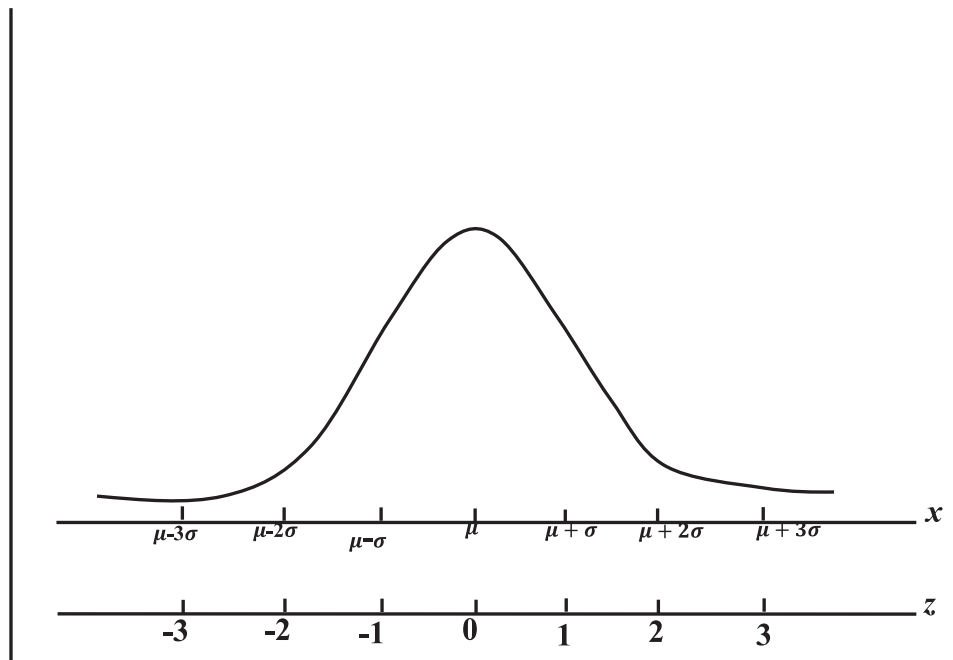


Fig. 27.3: Normal Distribution with Deviations from Mean

If x follows a normal distribution with mean as μ , and standard deviation σ : $x \sim N(\mu, \sigma)$ and z is another variate which is related to x such that

$z = \frac{x - \mu}{\sigma}$. Then z follows a normal distribution $z \sim N(0, 1)$ with 0 mean and 1

standard deviation. This simplest case of a normal distribution is known as the Standard Normal distribution. This is a special case when $\mu = 0$ and $\sigma = 1$

(Refer Fig 27.3). If $z = \frac{x - \mu}{\sigma}$ then $N(0, 1) = f(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}}; -\infty \leq z \leq +\infty$

Following results regarding the area under the Standard Normal distribution are useful:

Area between $z = \pm 3$ is 99.73%

Area between $z = \pm 2.58$ is 99 %

Area between $z = \pm 1.96$ is 95%

Let the ordinate of $F(p)$ be denoted by z_p , the value of the standard normal variate z such that the probability of its being exceeded is p i.e., $p(z > z_p) = p$. Area under the curve to the right of the ordinate at z_p is p .

Area of the right of $z = 1.645$ is 5% = $z_{.05}$

Area of the right of $z = 1.96$ is 2.5% = $z_{.025}$

Area of the right of $z = 2.33$ is 1% = $z_{.01}$

The approximate area under the standard normal curve is shown in Fig. 27.4.

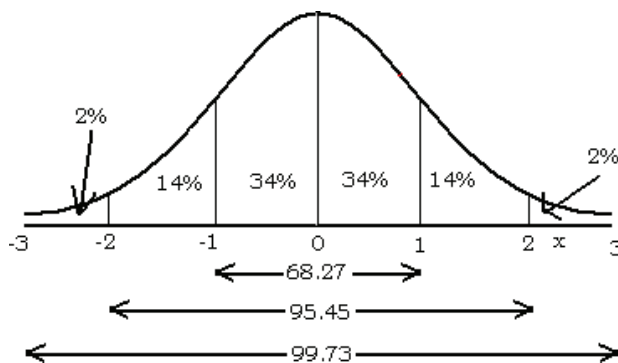


Fig 27.4: Area under the Standard Normal Curve

Example: The height distribution of a group of 10,000 men is normal with mean height 64.5" and standard deviation 4.5. Find the number of men whose height is

- less than 69" but greater than 55.5"
- less than 55.5"
- more than 73.5"

The mean and standard deviation of a normal distribution is given by

$\mu = 64.5''$ and $\sigma = 4.5$. We explain the problem graphically. As per Fig. 27.5, we can easily comprehend what we are asked to do. We are to find out the shaded regions but as we know area under a standard normal curve only, we have to reduce the given random variable into a standard normal variable.

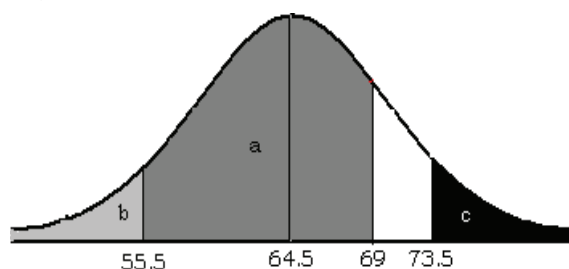


Fig 27.5: Standard Normal Distribution when $\mu = 64.5''$ and $\sigma = 4.5$.

Let X be the continuous random variable measuring the height of each man.

Therefore, $(X-64.5)/4.5 = z$ is a standard normal variable.

Table 27.6 shows values of z for corresponding values of x . In the table, true area under standard normal curve is given for only positive values of the standard normal variable and as the distribution is symmetrical, area under the curve for negative values of the standard normal variable is easy to find out.

Table 27.6: Values of z as per x value

X	Z
55.5	-2
64.5	0
69	1
73.5	2

- $P(55.5 < X < 69) = P(-2 < Z < 1) = \Phi(1) - \Phi(-2) = .82$, Therefore, men of height less than 69" but greater than 55.5" is $10000 \times 0.82 = 8200$
- $P(X < 55.5) = P(z < -2) = .02$, Therefore, men of height less than 55.5" is $10000 \times 0.02 = 200$
- $P(X > 73.5) = P(z > 2) = 1 - P(z < 2) = 1 - .98 = .02$, Therefore, men of height greater than 73.5" is $10000 \times 0.02 = 200$

For standard normal curve (say for the variable z) the area under the curve to the left is conventionally denoted by $\Phi(z_1)$. This is shown in Fig. 27.6.

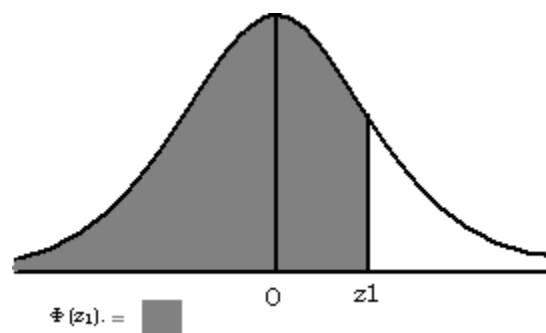


Fig 27.6: Area under the Standard Normal Curve denoted by $\Phi(z_1)$

Distribution of Standard Normal Variable

A standard Normal variable of a normal deviate is generally denoted by τ (or z), so that the p.d.f. of standard normal distribution may be written as

$$f(\tau) = \frac{1}{\sqrt{2\pi}} \exp(-\tau^2 / 2), -\alpha < \tau < \alpha$$

The distribution is symmetrical about 0. The p.d.f of standard normal distribution is

$$f(\tau) = \frac{1}{\sqrt{2\pi}} \exp(-\tau^2/2), -\alpha < \tau < \alpha$$

The properties of this distribution may be deduced from those of a general normal distribution. We shall denote by τ_α the value of τ such that

$$p[\tau > \tau_\alpha] = \alpha$$

It is called the upper α -point (or the upper 100 $\alpha\%$ point) of a standard variable. Because of the symmetry of the distribution about zero, we have

$$\tau_{1-\alpha} = -\tau_\alpha$$

Thus, the lower α point, $\tau_{1-\alpha}$ of a standard normal variable is the value of τ such that $p[\tau > \tau_{1-\alpha}] = \alpha$ which is the same as the upper α point in magnitude but has the opposite sign.

It follows from the theorem below that if x is normally distributed with mean μ and variance σ^2 , then $(x - \mu)/\sigma$ is a standard normal variable. Conversely, if $(x - \mu)/\sigma$ is a standard normal variable, then x is a normal variable with mean μ and variance σ^2 .

Theorem: If x is normally distributed with mean μ and variance σ^2 , then $y = a + bx$, where $b \neq 0$, is also normally distributed with mean $a + b\mu$ and variance $b^2\sigma^2$.

Proof: Let us denote the p.d.fs of x and y by $f(x)$ and $g(y)$, respectively.

Assuming $b > 0$, from the result if $y = a + bx$, then $\frac{y-a}{b} = x$

$$p[c < y < d] = p\left[\frac{c-a}{b} < x < \frac{d-a}{b}\right],$$

$$\int_c^d g(y) dy = \int_{(c-a)/b}^{(d-a)/b} f(x) dx = \int_c^d f\left(\frac{y-a}{b}\right) \frac{dx}{dy} dy \quad \dots\dots\dots(1)$$

If $b < 0$, we similarly have,

$$p[c < y < d] = p\left[\frac{d-a}{b} < x < \frac{c-a}{b}\right]$$

$$\int_c^d g(y) dy = -\int_c^d f\left(\frac{y-a}{b}\right) \frac{dx}{dy} dy \quad \dots\dots\dots(2)$$

Combining the two results (1 & 2), we get

$$g(y) = f\left(\frac{y-a}{b}\right) \left|\frac{dx}{dy}\right|$$

Note that

$$\frac{dx}{dy} = \frac{1}{\left(\frac{dy}{dx}\right)} = \frac{1}{b}$$

But

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{(x-\mu)^2}{2\sigma^2}\right]$$

$$\text{and } g(y) = \frac{1}{|b|\sigma\sqrt{2\pi}} \exp\left[-\frac{\left(\frac{y-a}{b}-\mu\right)^2}{2\sigma^2}\right]$$

Hence

$$= \frac{1}{|b|\sigma\sqrt{2\pi}} \exp\left[-\frac{(y-a-b\mu)^2}{2b^2\sigma^2}\right].$$

Check Your Progress 3

- 1) The mean weight of 500 students at a university is 151 lbs and the standard deviation is 15. Assuming the weights are normally distributed, find how many students:

- i) Weigh between 120 and 155 lbs.
- ii) Weigh More than 155 lbs.

Given: $\Phi(0.27) = 0.6064$; $\Phi(2.07) = 0.9808$; $\Phi(t)$ implies the area under the standard normal curve to the left of the ordinate at the point t .

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- 2) The mean of a normal distribution is 50 and 5% of the values are greater than 60. Find out the standard deviation of the distribution (given that the area under the standard normal curve between $z=0$ and $z=1.64$ is 0.45)

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